Shiyu Jiang

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GitHub: github.com/JasonJiangs | Google Scholar: citations 244; h-index 7; i10-index 8 | Last updated: April 3, 2026

RESEARCH INTERESTS

AI-driven Drug Discovery, Deep Learning Methods for Structural Biology, Large Language Models for Life Sciences.
Molecule Design and Optimization: Deep generative modeling for small molecules, RNA, and protein design.
Molecular Recognition: Geometric deep learning for binding affinity prediction, sequence-function modeling, and structure-aware learning.
Computational Biology: Single-cell foundation models, genomic language models, and data-driven biological discovery.

EDUCATION AND TRAINING

Ph.D. Student in Computational Biology and Bioinformatics *Aug. 2025 – May. 2026*
Dept. of Quantitative and Computational Biology, University of Southern California, Los Angeles, CA, USA
Selected coursework: QBIO 570 - Intro to Data Structures & Algorithms for Science; QBIO 540 - Prob & Stats for Computational Sciences; QBIO 578A & 578B - Compputational Molecular Biology; QBIO 502 - Molecular Biology for Quantitative Scientists;

MSE in Computer Science, Bioinformatics Track *Aug. 2022 – Dec. 2023*
Dept. of Computer Science, Johns Hopkins University, Baltimore, MD, USA
Selected coursework: CSCI 601.649 - Comp Genom: Applied Comparative Gen; CSCI 601.646: Sketching & Indexing for Sequences; Thesis Advisor: Prof. Kimia Ghobadi; Course Advisor: Prof. Ben Langmead

Committee: Kimia Ghobadi, Anton Dahbura, Claus Aranha (University of Tsukuba)
Thesis: Analyzing epidemic spread and emotion contagion with agent-based simulation and system dynamic modeling

B.S. in Computer Science; Minor in Mathematical Sciences *Aug. 2018 – May. 2022*
Dept. of Computer Science, Wenzhou-Kean University, Wenzhou, China
Advisor: Prof. Aloysius Wong

PEER-REVIEWED PUBLICATIONS

(† ‡ = equal contribution; * = co-corresponding author Google Scholar)

Representative Publications

- **Jiang S**†, Fang Z†, ... A roadmap for foundation models in single-cell omics. *In preparation*.
- **Jiang S***, Liu X, Wang JZ*. Evaluating DNA function understanding in genomic language models using evolutionarily implausible sequences. Revision in *Research journal*.
- **Jiang S**†, Taghavi A†, Wang T, Meyer SM, Childs-Disney JL, Li C, Disney MD*, Li Y*. Small Molecule Approach to RNA Targeting Binder Discovery (SMARTBind) Using Deep Learning Without Structural Input. Revision in *Nature press research journal*.
- Cheng L†, Zheng X†, **Jiang S**†, Hu Y, Liu Y, Yang K, Rui J, Ding H, Zhang M, Yuan T, Ye H, Li C, Yang KK, Huang X*, Xiao H*. Sequence Display enables large-scale sequence-activity datasets for rapid protein evolution. *Nature Biotechnology*, 2026. (AIP)
- Ding J†, Lin J†, **Jiang S**†, Wang Y, Mao Z, Fang Z, Tang J*, Li M*, Qiu X*. Tabula: A Tabular Self-Supervised Foundation Model for Single-Cell Transcriptomics. In *NeurIPS*, 2025.
- Su J†, He Y†, You S†, **Jiang S**, Zhou X, Zhang X, Wang Y, Su X, Tolstoy I, Chang X*, Lu H*, Yuan F*. A tri-modal protein language model enables advanced protein searches. *Nature Biotechnology*, 2025: 1–7.
- **Jiang S***, Kim H, Tanaka F, Aranha C*, Bogdanova A, Ghobadi K, Dahbura A. Simulating Disease Spread during Disaster Scenarios. In *Conference on Artificial Life (ALIFE)*, 2023.
- **Jiang S**, Abdalla H, Bi C, Zhu Y, Tian X, Yang Y*, Wong A*. HNOXPred: a web tool for the prediction of gas-sensing H-NOX proteins from amino acid sequences. *Bioinformatics*, 38(19): 4643–4644, 2022.

Under Review / In Preparation

- Cheng L†, Zheng X†, Ding H†, **Jiang S**, Hu Y, Wang C, Li C, Rui J, Ye H, Liu Y, Tian Z, Yuan T, Huang X*, Xiao H*. Sequence Display generates large-scale sequence-activity datasets for reprogramming SlugCas9 toward difficult-to-access PAMs. Under review by *Cell press research journal*.
- Ding J†, Lin J†, Mechtel N†, Mao Z†, Lu R†, **Jiang S**, ..., Wang Y, Fang Z, Ouyang W, Tang J*, Li M*, Qiu X*. Privacy-preserving foundation model with tabular learning for predictive single-cell perturbation in disease and aging. Under review by *Nature press research journal*.
- Zhou X†, Han C†, Zhang Y†, Du H†, Tian J†, Su J‡, Liu R‡, Zhuang K‡, **Jiang S**‡, Gitter A, Liu L, Li H, Wu M, You S, Yuan Z, Ju F, Zhang H, Zhang W, Dai F, Zhou Y, Tao Y, Wu D, Zhao Z, Liu Y*, Lu H*, Yuan F*. Decoding the Molecular Language of Proteins with Evolla. Under review by *Nature press research journal*.

Journal Articles

- Yuan T†, Zhang M†, Cheng L, Zheng X, **Jiang S**, Huang X, Xiao H*. Biocatalytic Synthesis of N-protected α-Amino Acids through 1,3-Nitrogen Migration by Nonheme Iron Enzymes. *Journal of the American Chemical Society*, 2025.

- Su J, Li Z, Han C, Zhou Y, He Y, Shan J, Zhou X, Chang X, **Jiang S**, Ma D, OPMC, Steinegger M, Ovchinnikov S, Yuan F*. Democratizing protein language model training, sharing and collaboration. *Nature Biotechnology*, 2025: 1–7.
- Hu Y, Wang Y, Cheng L, Wang C, Liu Y, Wang Y, Chen Y, Yang S, Guo Y, **Jiang S**, Yang K, Xiao H*. Engineering unnatural cells with a 21st amino acid as a living epigenetic sensor. *Nature Communications*, 16(1): 9388, 2025.
- Hu Y†, Cheng L†, Liu Y, Liu R, **Jiang S**, Yuan T, Wang Y, Ye H, Xiao H*. Biosynthesis of Unnatural Cyclodipeptides through Genetic Code Expansion and Cyclodipeptide Synthase Evolution. *Journal of the American Chemical Society*, 147(38): 34517–34526, 2025.
- Ye H, **Jiang S**, Yan Y, Zhao B, Grant ER, Kitts DD, Yada RY, Singh AP, Baldelli A, Yang T*. Integrating Metal Phenolic Networks-Mediated Separation and Machine Learning-Aided SERS for High-Precision Quantification and Classification of Nanoplastics. *ACS Nano*, 18(38): 26281–26296, 2024. [Cover article].
- Guo Y†, Cheng L†, Hu Y, Zhang M, Liu R, Wang Y, **Jiang S**, Xiao H*. Biosynthesis of halogenated tryptophans for protein engineering using genetic code expansion. *ChemBioChem*, 25(20): e202400366, 2024. [VIP paper].

Conference Articles

- Zheng X, **Jiang S**, Seabra G, Li C, Li Y*. Apo2Mol: 3D Molecule Generation via Dynamic Pocket-Aware Diffusion Models. In *AAAI*, 2026.
- Qi X†, Zhang Z†, Handoko AB†, Zheng H, Chen M, Huy TD, Ohan VMH, Zhang L, Cheng L, **Jiang S**, Zhang Z, Liao Z, Zhao Y*, To MS*. ProjectedEx: Enhancing Generation in Explainable AI for Prostate Cancer. In *IEEE CBMS*, 2025.
- Song Z†, Ouyang G†, Fang M†*, Na H, Shi Z, Chen Z, Fu Y, Zhang Z, **Jiang S**, Fang M, Chen L, Chen X*. Hazards in Daily Life? Enabling Robots to Proactively Detect and Resolve Anomalies. In *NAACL*, 2025.
- Zheng S, Wu Y, **Jiang S**, Lu C, Gupta G*. Deblur-yolo: Real-time object detection with efficient blind motion deblurring. In *IJCNN*, 2021.

Workshop Papers

- **Jiang S***, Liu X, Wang JZ*. Predicting function of evolutionarily implausible DNA sequences. In *ICML 2025 Workshop on Generative AI and Biology*.

PROFESSIONAL EXPERIENCE

Remote Research Assistant, Dept. of Chemistry, Rice University *Mar. 2024 – Aug. 2025*
Advisor: Prof. Han Xiao & Dr. Kevin K. Yang (Microsoft Research)

- Assisted protein evolution by fine-tuning protein language models on sequence display assay data.
- Analyzed multiple assay datasets, including next-generation sequencing and nanopore sequencing data.

Remote Research Assistant, Dept. of Medicinal Chemistry, University of Florida *Aug. 2023 – Jul. 2025*
Advisors: Prof. Yanjun Li & Prof. Matthew D. Disney (Scripps Research)

- Developed SMARTBind, a model for RNA–ligand interaction prediction, enabling the identification of novel small molecules targeting the precursor of oncogenic microRNA-21, with hits validated by *in vitro* and cellular assays.
- Developed a diffusion-based molecular generation model that considers the dynamic induced fit of protein pockets.

Research Associate, Westlake University *Aug. 2024 – Jul. 2025*
Advisors: Prof. Fajie Yuan & Prof. Zitong Jerry Wang

- Developed and evaluated large-scale protein language models for downstream application and wet lab discovery.
- Evaluated the evolutionary understanding of genomic language model by proposing a dataset of mutated synthetic expression cassettes.

Visiting Master Student, Dept. of Genetics, Stanford University *Aug. 2023 – Jul. 2025*
Advisor: Prof. Xiaojie Qiu & Dr. Jiayuan Ding

- Pretrained a single-cell foundation model, Tabula, using Tabular learning and federated learning.
- Designed an *in silico* perturbation experiment to predict gene regulation across developmental biological systems including Hematopoiesis, Cardiogenesis, Neurogenesis, and Pancreatic Endogenesis under zero-shot settings.
- Benchmarked with scGPT, Geneformer, and scFoundation on multiple cell-level and gene-level downstream tasks.

Lab Specialist, School of Medicine, University of Virginia *Jan. 2024 – Jul. 2024*
Advisors: Prof. Chongzhi Zang & Prof. Sheng'en Shawn Hu

- Developed and evaluated a ChIP-seq broad peak-calling tool (SICER).

Postgraduate Research Assistant, Dept. of Computer Science, Johns Hopkins University *Oct. 2022 – Jul. 2023*
Advisors: Prof. Kimia Ghobadi, Prof. Anton Dahbura & Prof. Claus Aranha (Univ. of Tsukuba)

- Developed an agent-based modular simulator for small-community disease spread.

Software Engineer Intern, Alibaba Cloud, Hangzhou, China *Jun. 2022 – Aug. 2022*
Mentor: Mr. Jiabang Pan

- PolarDB database development (Java and MySQL).

Undergrad Research Assistant, Dept. of Biology, Wenzhou-Kean University *Oct. 2021 – May. 2022*
Advisor: Prof. Aloysius Wong

- Bioinformatics webserver development for protein motif recognition.

POSTERS & PRESENTATIONS

- "Small Molecule Approach to RNA Targeting Binder Discovery (SMARTBind) Using Deep Learning Without Structural Input." Poster presentation, UF Health Cancer Institute Annual Research Showcase 2026, University of Florida, Gainesville, FL, USA. Jan. 29, 2026.
- "Tabula: A Tabular Self-Supervised Foundation Model for Single-Cell Transcriptomics." Poster presentation, *NeurIPS* 2025, Hilton Reforma and San Diego Convention Center, San Diego, CA, USA. Dec. 2, 2025.
- "Predicting function of evolutionarily implausible DNA sequences." Poster presentation, Q-BIO 2025 Conference: Emergent Orders in Living Systems Across Scales, Center of Quantitative Biology, Peking University, Beijing, China. Jul. 23, 2025.
- "Predicting function of evolutionarily implausible DNA sequences." Poster presentation, *ICML* 2025 Workshop for Generative AI and Biology, Vancouver, Canada. Jul. 18, 2025.
- "Sequence Display-Enable Machine Learning for Protein Evolution." Poster presentation, Synthetic Biology: Engineering, Evolution, & Design, Houston, TX. Jun. 23, 2025.
- "Simulating Disease Spread during Disaster Scenarios." Talk presentation, The 2023 Conference on Artificial Life, Sapporo, Japan. Jul. 23, 2023.

PROFESSIONAL SERVICE

Journal Reviewer

- *IEEE Transactions on Computational Biology and Bioinformatics*
- *PLoS Computational Biology*

Conference Reviewer

- *Association for the Advancement of Artificial Intelligence (AAAI) 2026*

HONORS & AWARDS

USC Dana and David Dornsife College Graduate School Fellowship *2025*